Mock 2 M01

Delivery Quality - Lakeflow Spark Declarative Pipeline

SUBMISSION TEMPLATE

Difficulty	Marks	Tasks	Screenshot rule
Medium	20	5	Multiple allowed per task

Trainee: _____Soumiya Ayyappan_____ Candidate ID: _____10877547_____ Catalog: _____cat_ltm10877547_____

Objective

Build a triggered declarative Bronze streaming ingestion with filtered Silver materialized view and a materialized hub-level summary.

Evidence policy

SCREENSHOT POLICY: There is NO one-task/one-screenshot limit. For every assessed task, provide as many readable images as needed to establish ALL listed rubric criteria. Where the rubric tests implementation AND its effect, supply both code/configuration screenshots and result screenshots. If code is long, use consecutive clearly labelled Code A, Code B, Code C images; do not shrink or stitch until unreadable. Label images Q_T01_Code_A, Q_T01_Result_A, etc. Keep table/pipeline names, executed statements and relevant returned rows visible. An output screenshot cannot replace required implementation evidence, and code alone cannot replace required output evidence. Never rerun MERGE, DELETE or other mutations solely to capture evidence: capture prior run/history and current read-only validations.

Environment and object names

Use your existing assigned Unity Catalog and schema mock2. Use the approved existing external location; the instructor supplies your trainee-specific ADLS URI. Create a question-specific EXTERNAL volume at that full child URI. Do not create a new catalog, storage credential or external location. Upload original CSV filenames into input/ under the question volume. Store managed Delta tables and pipeline outputs in schema managed storage, not in the volume.

Setting	Required value
Catalog	Your assigned existing catalog
Schema	mock2
External volume	m2_m01_files
Child ADLS path	mock2/m01/files/ beneath your assigned trainee root
Input folder	/Volumes/<catalog>/mock2/m2_m01_files/input/
Required output objects	m2_m01_delivery_pipeline, m2_m01_delivery_bronze, m2_m01_delivery_valid, m2_m01_hub_summary

Key terms

- Bronze: retains all input rows, without applying delivery status filters

- Silver: keeps only Delivered rows with positive delivery_fee

- materialized view: pipeline-managed saved result refreshed by a pipeline update

Source dataset and column dictionary

Provided file(s): M01_delivery_events.csv

Column	Required type	Use in operations
delivery_id	STRING	Unique delivery ID
hub	STRING	Delivery hub
distance_km	INTEGER	Route distance
delivery_fee	INTEGER	Fee
status	STRING	Delivery state

Task requirements and evidence

Task 1. Prepare governed input and pipeline | 3 marks

1.1 Create the question external volume and upload the single CSV into input/.

1.2 Create or configure the assigned triggered Lakeflow SDP pipeline. Select SQL pipeline source and the existing catalog/mock2 destination.

1.3 Attach the actual pipeline source file to the pipeline definition.

Expected result: Named input and triggered SQL pipeline; correct destination and attached source.

Evaluator evidence - every item below must be legible:

A. Setup UI: EXTERNAL volume details and source file.

B. Config UI: pipeline name, triggered mode, catalog/schema and source association.

Use filenames such as M2_M01_T01_Code_A.png, M2_M01_T01_Result_A.png or M2_M01_T01_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.

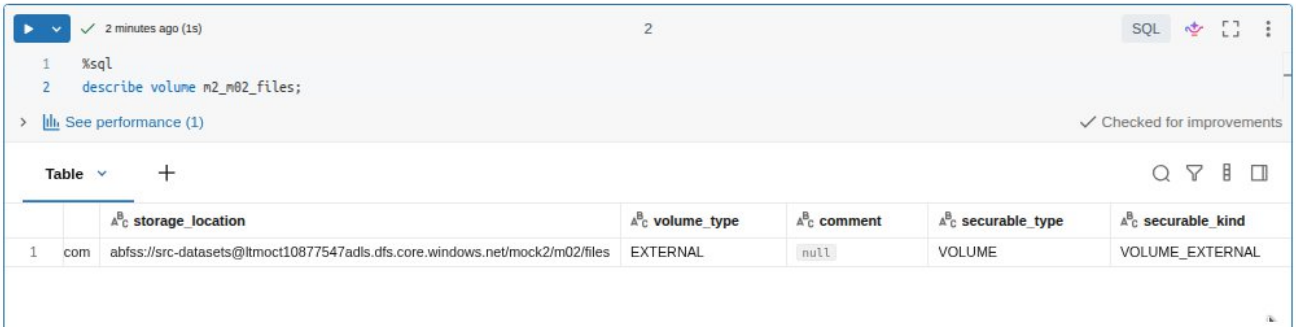


	current_catalog()	current_schema()
1	cat_ltm10877547	mock2

1 row | 17.608s runtime

Refreshed at 10/27/26, 12:53 PM (47m ago)

This result is stored as `_sqldf` and can be used in other Python and SQL cells.



2 minutes ago (1s)

SQL

```
%sql
describe volume m2_m02_files;
```

See performance (1)

Checked for improvements

Table

		storage_location	volume_type	comment	securable_type	securable_kind
1	com	abfss://src-datasets@ltmoct10877547adls.dfs.core.windows.net/mock2/m02/files	EXTERNAL	null	VOLUME	VOLUME_EXTERNAL

[Paste readable screenshot here]

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Task 2. Define Bronze streaming ingestion | 4 marks

2.1 Define a Bronze streaming table that reads the input CSV using the supported pipeline file-ingestion mechanism.

2.2 Assign the five dataset column types and keep all eight rows, including Pending and Cancelled.

2.3 Do not exclude invalid business states at Bronze.

Expected result: Bronze table has 8 rows, 5 source columns.

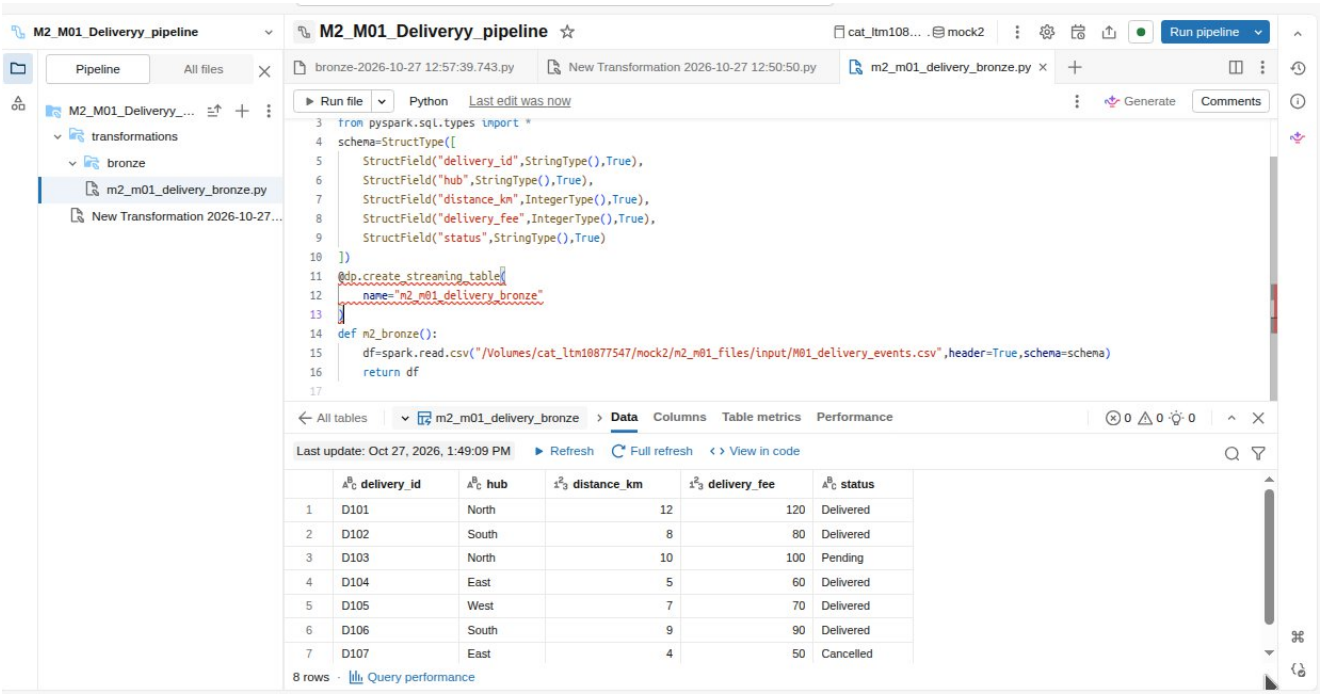
Evaluator evidence - every item below must be legible:

A. Code A/B: full Bronze definition and file-source settings, including all typed columns.

B. Result (after successful update): named Bronze dataset and count 8.

Use filenames such as M2_M01_T02_Code_A.png, M2_M01_T02_Result_A.png or M2_M01_T02_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.



[Paste readable screenshot here]

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Task 3. Define valid-delivery Silver | 4 marks

3.1 Create a materialized view reading the Bronze dataset.

3.2 Retain only status equal to Delivered AND delivery_fee greater than zero.

3.3 Ensure all five input columns remain present.

Expected result: Silver has 6 rows; excludes D103 Pending and D107 Cancelled.

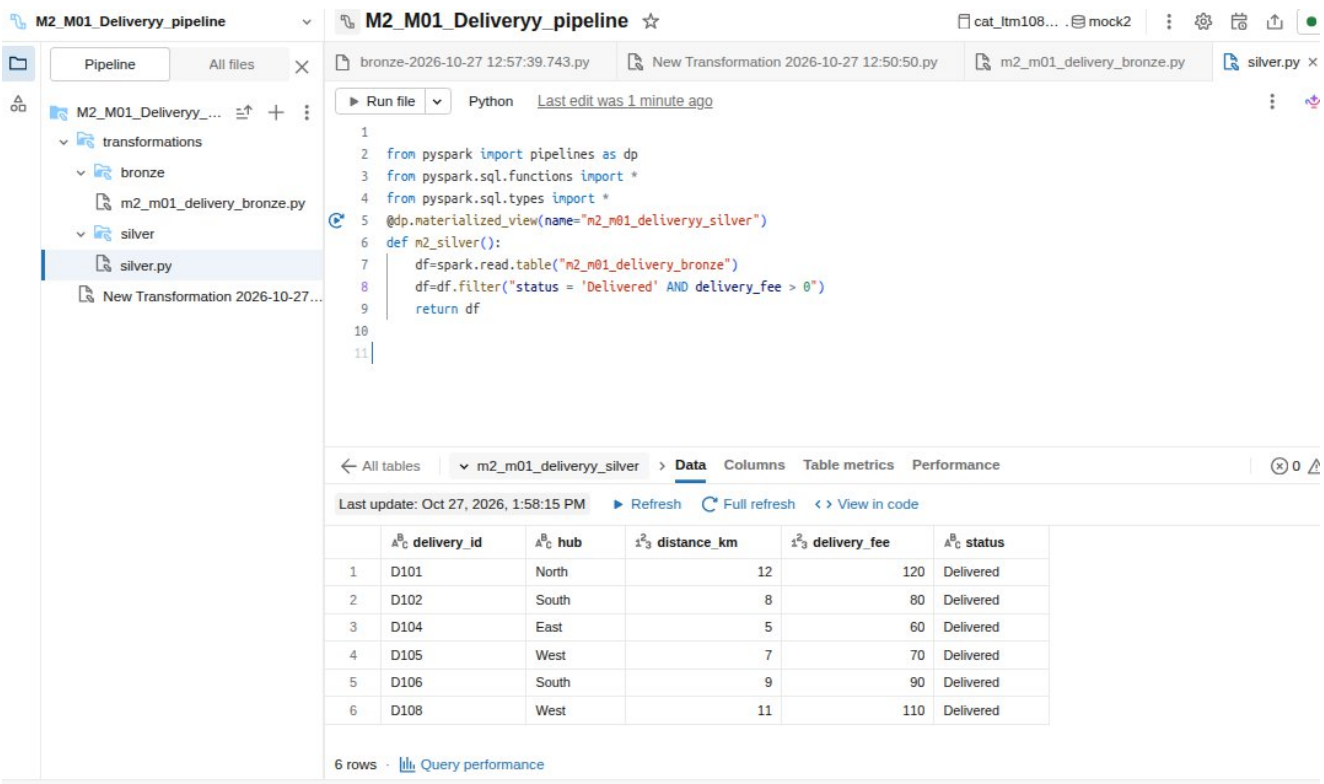
Evaluator evidence - every item below must be legible:

A. Code: Silver view definition with Bronze dependency and both filters.

B. Result (after successful update): count 6 and proof excluded IDs are absent.

Use filenames such as M2_M01_T03_Code_A.png, M2_M01_T03_Result_A.png or M2_M01_T03_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.



[Paste readable screenshot here]

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Task 4. Create hub-level fee metric | 4 marks

4.1 Create a materialized hub summary derived from the saved Silver view.

4.2 Group by hub; calculate delivery_count as number of valid deliveries and total_fee as sum of delivery_fee.

4.3 Order the four hubs for readable final reporting.

Expected result: East 1/60; North 1/120; South 2/170; West 2/180.

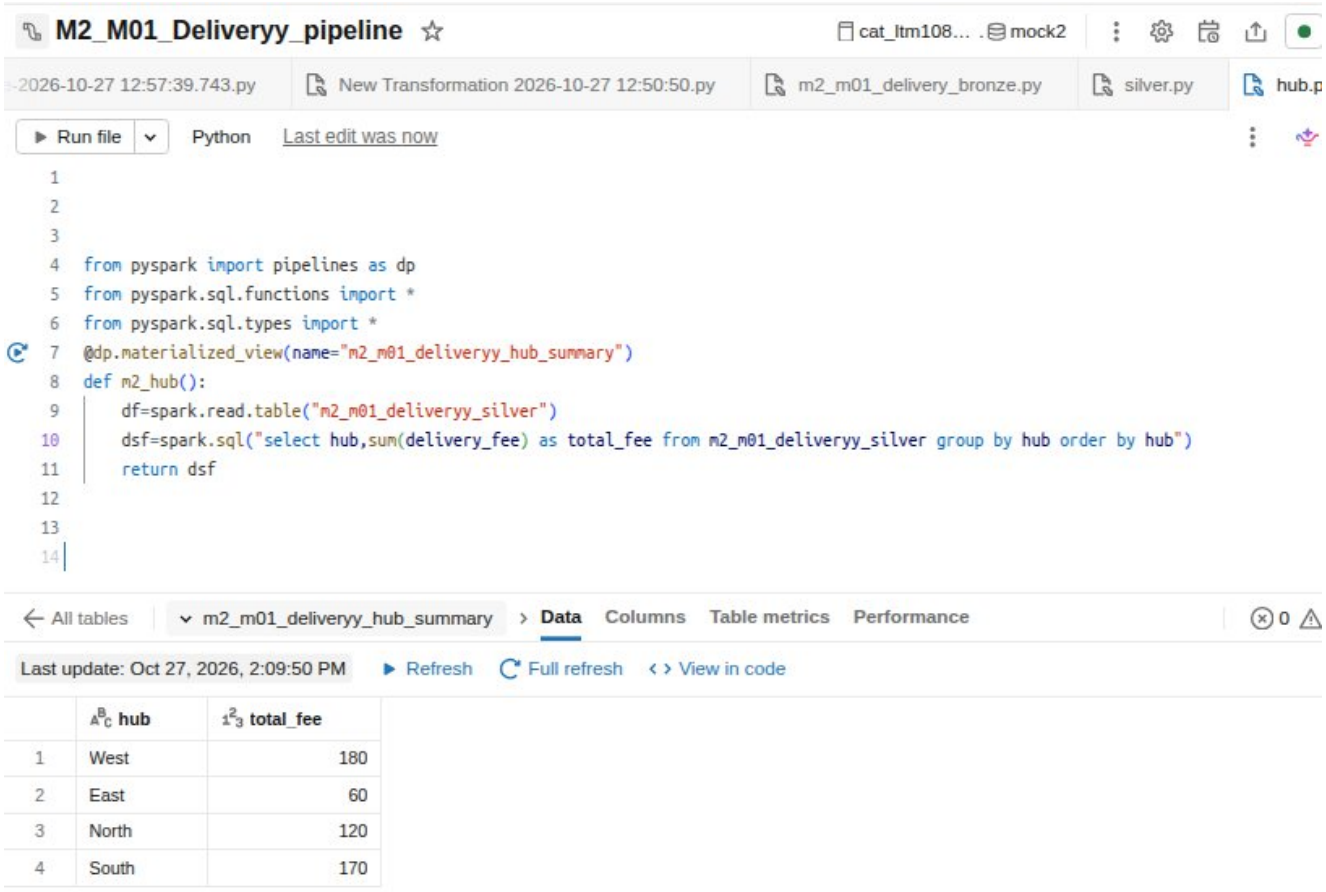
Evaluator evidence - every item below must be legible:

A. Code: complete group-by definition, count and sum expressions.

B. Result: four ordered hub rows with hub, delivery_count, total_fee.

Use filenames such as M2_M01_T04_Code_A.png, M2_M01_T04_Result_A.png or M2_M01_T04_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.



[Paste readable screenshot here]

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Task 5. Run and reconcile pipeline output | 5 marks

5.1 Run a triggered pipeline update once; ensure the pipeline completes successfully.

5.2 Inspect graph dependencies Bronze -> Silver -> hub summary and destination catalog/schema.

5.3 Reconcile saved Bronze, Silver and hub row counts and total delivered fee.

Expected result: Successful pipeline update; Bronze 8, Silver 6, summary 4; total delivered fee 530.

Evaluator evidence - every item below must be legible:

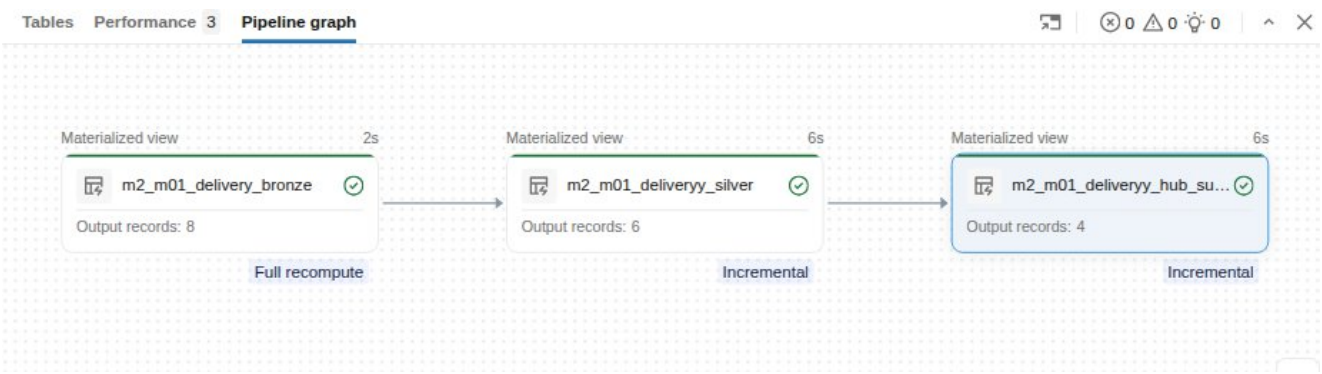
A. UI: successful update status and all three named nodes in dependency graph.

B. Results: counts 8/6/4 and total delivery_fee 530.

C. If needed, separate evidence of configured destination, saved output identity and latest run.

Use filenames such as M2_M01_T05_Code_A.png, M2_M01_T05_Result_A.png or M2_M01_T05_UI_A.png. Add _B / _C where required. These are naming examples, NOT a fixed screenshot count.

SCREENSHOT PLACEHOLDER - paste the required task evidence below; duplicate this holder for additional screenshots.



[Paste readable screenshot here]

Additional screenshots (Code B / Result B / UI B, etc.) may be inserted after this placeholder.

Evidence reference / workspace object: _____

Expected final results

Task 1: Named input and triggered SQL pipeline; correct destination and attached source.

Task 2: Bronze table has 8 rows, 5 source columns.

Task 3: Silver has 6 rows; excludes D103 Pending and D107 Cancelled.

Task 4: East 1/60; North 1/120; South 2/170; West 2/180.

Task 5: Successful pipeline update; Bronze 8, Silver 6, summary 4; total delivered fee 530.

Evaluation rubric and evidence map

Task	Rubric metric - evidenced by screenshots	Marks
1	Setup UI: EXTERNAL volume details and source file.	2
1	Config UI: pipeline name, triggered mode, catalog/schema and source association.	1
2	Code A/B: full Bronze definition and file-source settings, including all typed columns.	2
2	Result (after successful update): named Bronze dataset and count 8.	2
3	Code: Silver view definition with Bronze dependency and both filters.	2
3	Result (after successful update): count 6 and proof excluded IDs are absent.	2
4	Code: complete group-by definition, count and sum expressions.	2
4	Result: four ordered hub rows with hub, delivery_count, total_fee.	2
5	UI: successful update status and all three named nodes in dependency graph.	2
5	Results: counts 8/6/4 and total delivery_fee 530.	2
5	If needed, separate evidence of configured destination, saved output identity and latest run.	1
TOTAL	No hidden checks or extra requirements	20

Submission requirements

Submit this matching template populated with readable taskwise screenshots. Provide 2 or 3 screenshots within a task when the implementation is long, or when code/configuration and results cannot both fit legibly. Show every assessed rubric metric, not just a success message. No fixed screenshot count, no scripts supplied in assessment instructions. Keep the executed workspace artifacts and history available. Do not rerun data-changing operations to recapture evidence.